

# Diego De Pablo

## Software Engineer | Full-Stack & Data Systems

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### Professional Summary

Graduate in **Bioinformatics Engineering** and current **M.S. in Computer Engineering** candidate. Specialized in **full-stack software engineering**, **cloud-native architectures**, and **data-driven AI systems**. Proven expertise building high-performance APIs with **FastAPI**, modern frontends with **React/Svelte**, container orchestration on **Kubernetes**, and integrating **LLM agents** for federated data spaces.

### Professional Experience

Software Engineer, **Khaos Research**

Sept 2025 – Present

- End-to-end full-stack development for federated, interoperable data spaces (SEDIA project).
- Architected an advanced LLM-powered semantic search engine for natural language querying and automated dataset discovery.
- Engineered modular metadata extraction services (+30 scientific formats) and autonomous LLM analysis agents.

FullStack Developer, **Msurgery**

March 2025 – July 2025

- Spearheaded structural migration of core repositories from Svelte 4 to Svelte 5, significantly optimizing run-time reactivity.
- Developed core features for the SMOTTS medical platform and digital twin simulation models.
- Managed international stakeholder communications and delivered the technical showcase at Polo Digital.

Junior Software Engineer, **T&B Company Group**

June 2021 – December 2021

- Modernized mission-critical services into Java and Angular frameworks; diagnosed and fixed production issues.
- Authored technical REST API specifications and comprehensive documentation to accelerate developer onboarding.

### Featured Projects (Interactive Portfolio)

🔗 IoT Patient Monitoring Bracelet (Undergraduate Thesis)

C++, FreeRTOS, MQTT, Flutter, FastAPI

- Embedded smart wearable and real-time biomedical telemetry streaming system with anomaly detection and clinical alerts.
- Awarded **Grade 10/10 with Honors distinction**. Portfolio link includes interactive architecture diagram and demo.

🔗 Cloud-Native Kubernetes & GitOps Platform

K8s, Argo CD, Helm, Envoy, Longhorn

- Production-grade 7-stage learning and deployment lab: GitOps automation with Argo CD, Sealed Secrets, and distributed block storage.

🔗 TITAN Workflow Agent

FastAPI, LangChain, LLM Agents, Docker

- Autonomous AI agent for validated planning and execution of scientific bioinformatics pipelines over federated data.

### Education & Languages

B.S. IN HEALTH ENGINEERING (BIOINFORMATICS)

GPA: 7.96/10 Sep 2021 – Jul 2025

- Highest Academic Honors in **Programming Fundamentals**, **Intelligent Systems**, and **Bioinformatics Projects**.

M.S. IN COMPUTER ENGINEERING

2026 – Present

Languages: **Spanish** (Native) · **English** (B2 / Full professional and technical proficiency).

### Technical Skills

**Languages:** Python, TypeScript, JavaScript, SQL, C++, C, C#, Java, R.

**Backend & Cloud:** FastAPI, Node.js, Express, Docker, Kubernetes, Helm, Argo CD, CI/CD, Linux.

**Frontend:** React, Svelte / SvelteKit, Angular, HTML5, CSS3, Tailwind CSS.

**Data & Artificial Intelligence:** LangChain, OpenAI / Ollama APIs, Polars, Pandas, scikit-learn, PyTorch.